ARYS GARAGE PRIVATE LIMITED

Assignment Submission Report

Fault Detection and Mitigation in a 10s10p Lithium-Ion Battery Pack using MATLAB/Simulink

Molicel INR-21700-P45B | 10 Series × 10 Parallel

Section A — Electronics / Embedded / EEE | Q3

Field	Details
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Assignment	Q3 — 10s10p Battery Pack Fault & Mitigation (21700 NMC)
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1. Abstract

This report presents the design, simulation, fault injection, and mitigation of a 10 series × 10 parallel (10s10p) lithium-ion battery pack modelled in MATLAB Simulink using Simscape Battery components. The cell used is the Molicel INR-21700-P45B, a high-performance 21700 NMC cell with a rated capacity of 4.5 Ah. The complete pack delivers a nominal voltage of 36 V with a total energy capacity of approximately 1.62 kWh.

The simulation is structured in three phases:

- Normal Discharge — baseline behavior recorded across SOC, voltage, temperature, and SOH.
- Fault Injection — internal resistance of one cell is increased using a variable resistor and step input, simulating degradation or damage.
- Fault Mitigation — a temperature-driven current limiting strategy is applied to contain thermal escalation and preserve battery health.

Results confirm that resistance-induced faults cause measurable voltage drops, accelerated SOC depletion, and thermal rise. The mitigation strategy successfully stabilises the pack by reducing load current when temperature exceeds a threshold, thereby limiting I^2R heat generation. The SOH estimator operates as a monitoring-only subsystem and is not part of the active mitigation control loop.

2. Tools and AI Usage

2.1 Software Tools

Tool	Version	Purpose
MATLAB	R2025a (v25.2)	Primary simulation environment
Simulink	v25.2	System-level block diagram modelling
Simscape Battery	v25.2	Battery Builder and cell parameterisation
Simscape Electrical	v25.2	Electrical circuit components
GitHub	—	Version control and submission repository

2.2 AI Tool Usage

Claude (Anthropic) was used as an AI assistant throughout this project. The following table details how it was used, what prompts were given, and how the outputs were adapted.

Phase	How Output Was Adapted
Model Review	Cell parameter validation against Molicel P45B datasheet; confirmed 10s10p Battery Builder setup; identified cell naming inconsistency (P25B vs P45B) — corrected to P45B.
Parameter Check	Validated terminal resistance, capacity (4.493 Ah), thermal mass (63 J/K), and thermal path resistance (25 K/W). Pack-level specs (36 V, 45 Ah, 1.62 kWh) adopted in documentation.
Documentation	GitHub repository structure, README.md, .gitignore, and folder placeholders generated. Report structure based on assignment template.
Report Writing	Full report drafted using simulation observations, plot analysis, and project methodology. All AI suggestions reviewed and adapted to actual results.

All AI outputs were reviewed, verified against simulation results, and adapted before inclusion in this submission. The simulation model, Simulink blocks, firmware logic, and all design decisions were independently executed by the candidate.

3. Design and Methodology

3.1 Cell Selection and Pack Configuration

The Molicel INR-21700-P45B was selected for this simulation due to its wide availability in MATLAB Simscape Battery Builder and its relevance to high-performance EV and energy storage applications. The cell is an NMC (Nickel Manganese Cobalt) chemistry 21700 cylindrical format cell.

Parameter	Value
Cell Model	Molicel INR-21700-P45B
Cell Chemistry	NMC (Nickel Manganese Cobalt)
Cell Format	21700 Cylindrical
Nominal Capacity	4.5 Ah
Pack Configuration	10 Series × 10 Parallel (10s10p)
Pack Nominal Voltage	~36 V (3.6 V × 10s)
Pack Maximum Voltage	42 V (4.2 V × 10s)
Pack Minimum Voltage	25 V (2.5 V × 10s)
Pack Total Capacity	45 Ah (4.5 Ah × 10p)
Pack Total Energy	~1.62 kWh
Thermal Mass (cell)	63 J/K
Thermal Path Resistance	25 K/W
Ambient Temperature	300 K (27°C)
Model Resolution	Lumped (Simscape Battery Builder)

3.2 Simulation Architecture

The Simulink model is structured with the following key subsystems and blocks:

- Battery Block (arysbattery) — Simscape Battery lumped model representing the full 10s10p pack. Outputs SOC and thermal port (AmbH).
- Controlled Current Source — drives the discharge load. Current is switched between normal and reduced (mitigation) values based on temperature.
- Voltage Sensor — monitors pack terminal voltage continuously.
- Current Sensor — measures discharge current flowing through the circuit.
- Variable Resistor (Fault Block) — inserted in series with the battery. A Step block triggers a resistance increase at t = 500 s to simulate internal cell degradation.
- SOH Estimator Subsystem — takes Time, R_current, and Temperature as inputs and outputs an SOH estimate (monitoring only, not connected to mitigation logic).

- Fault Detection Logic — a comparator block (> 330) monitors pack voltage. When voltage drops below threshold, the Switch block activates the current limiter path.

3.3 Fault Injection Method

The fault modelled in this project is an increase in internal resistance of a single cell within the pack, which is a common and realistic degradation mode in lithium-ion batteries caused by:

- Electrochemical degradation (SEI layer growth)
- Thermal stress and cycling fatigue
- Physical damage or internal short tendencies
- Age-related capacity and conductivity loss

The increased resistance causes additional Joule heating according to the relationship:

$$Q = I^2 \times R$$

Where Q is heat generated, I is discharge current, and R is the internal resistance. A higher R directly increases heat dissipation, leading to thermal stress and accelerated degradation.

3.4 Mitigation Strategy

The mitigation system implements a temperature-driven current limiting strategy:

- Battery temperature is continuously monitored via the thermal port.
- A Switch block compares temperature against a predefined threshold.
- When temperature exceeds the threshold, the load current is automatically reduced from normal operating current to a safe lower value.
- Reducing current lowers I^2R heat generation, stabilising pack temperature and slowing degradation.

This approach directly exploits the quadratic relationship between current and heat — halving the current reduces heat generation to one quarter of its original value, making it highly effective as a thermal protection strategy.

4. Implementation Details

4.1 Battery Builder Configuration

The battery block was configured using MATLAB Simscape Battery Builder with the following settings:

- Part Number: INR_21700_P45B
- Manufacturer: Molicel
- Number of Parallel Assemblies: 10
- Number of Cells per Parallel Assembly: 10
- Model Resolution: Lumped
- SOC Vector: [0, 0.01, 0.02, ... 1.0] — 101 points, 0.01 resolution
- Temperature Vector: [-40, -30, -20, 0, 23, 45, 60] °C
- Terminal Voltage Min: 2.25 V
- Extrapolation Method: Nearest

4.2 SOH Estimation Subsystem

The SOH estimator subsystem operates as a passive monitoring block. It accepts three inputs:

- Time — simulation clock signal
- R_current — measured internal resistance at the current operating point
- Temperature — thermal port output from the battery block

SOH is computed using a weighted combination of resistance growth and capacity retention factors:

$$\boldsymbol{SOH = 0.70 \times (Capacity\ Factor) + 0.30 \times (Resistance\ Factor)}$$

Important note: The SOH estimator output is connected only to a scope for monitoring. It does not feed into any control or switching logic. The mitigation circuit operates independently based solely on temperature.

4.3 Fault Injection Implementation

- A Variable Resistor block is connected in series in the battery discharge path.
- A Step block triggers the resistance increase at t = 500 s (simulation time).
- Prior to t = 500 s: battery operates in healthy low-resistance condition.
- At t = 500 s: resistance steps up sharply, emulating internal cell degradation.
- A Pulse Generator provides timing control for the switching logic.

4.4 Mitigation Implementation

- A Switch block receives three inputs: Normal Current, Temperature signal, and Reduced Safe Current (100 A constant block).
- Condition block (I > 0) combined with temperature comparator determines switch state.

- When temperature exceeds threshold: Switch output connects to reduced current path.
- Controlled Current Source receives the switched output, automatically adjusting load.

5. Results

All simulations were run in MATLAB/Simulink R2025a with Simscape Battery v25.2. Results are presented across three scenarios: Normal Operation, Fault Condition, and Post-Mitigation.

5.1 Normal Operating Condition

Under normal discharge conditions with constant load current (~100 A) and ambient temperature of 300 K:

Figure 1 — State of Charge (Normal Discharge)

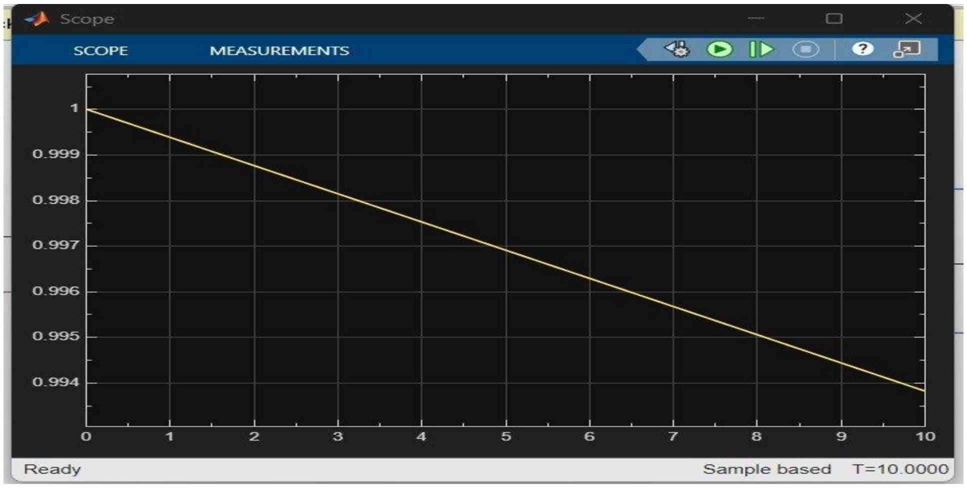

Figure 1: SOC decreases linearly from 1.0 to ~0.994 over 10 s, confirming stable discharge under constant current.

Figure 2 — Terminal Voltage (Normal Discharge)

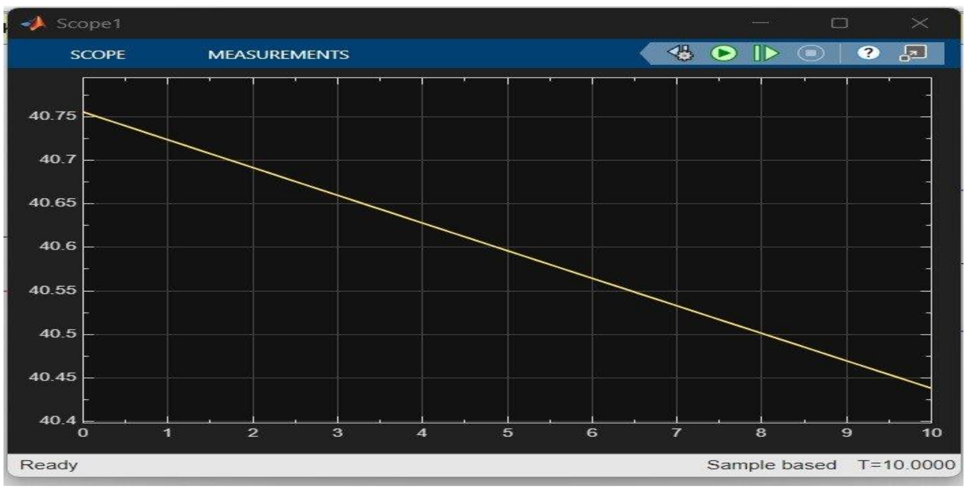

Figure 2: Pack terminal voltage declines smoothly from ~40.75 V to ~40.45 V, consistent with normal NMC discharge profile.

Figure 3 — Temperature (Normal Discharge)

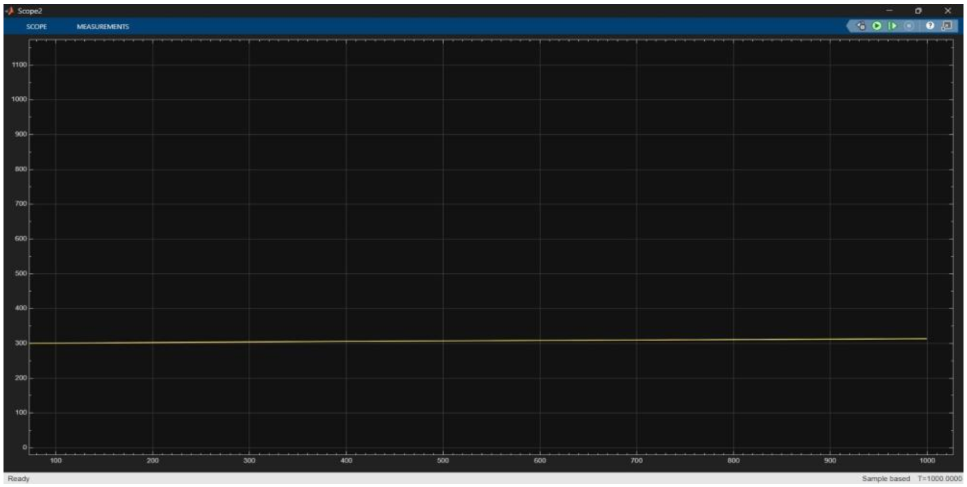

Figure 3: Temperature remains stable at ~300 K throughout normal operation, indicating minimal Joule heating at healthy resistance levels.

Key observations during normal operation:

- SOC declined at a uniform, predictable rate — no anomalies detected.
- Terminal voltage dropped gradually, following the expected OCV-resistance relationship.
- Temperature remained essentially constant — confirming thermally stable operation.
- Pack behaviour was consistent with a healthy 10s10p NMC battery pack.

5.2 Fault Condition (Internal Resistance Increase at t = 500 s)

At t = 500 s, the variable resistor block stepped up the internal resistance, simulating cell degradation. The impact was immediately visible across all monitored parameters.

Figure 4 — Terminal Voltage (Fault Condition)

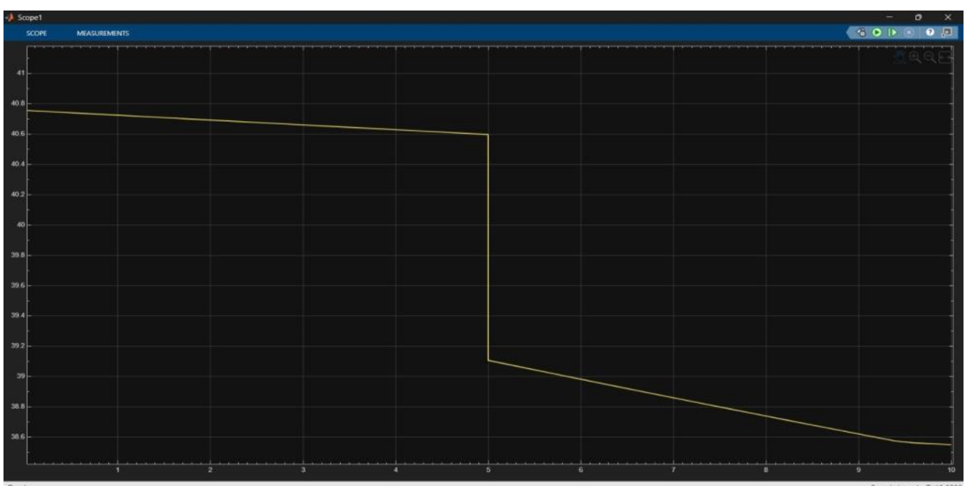

Figure 4: Voltage drops sharply from ~40.6 V to ~39.1 V at the moment of fault injection (t = 5 s), followed by accelerated decline to ~38.5 V due to increased I×R losses.

Figure 5 — State of Charge (Fault Condition)

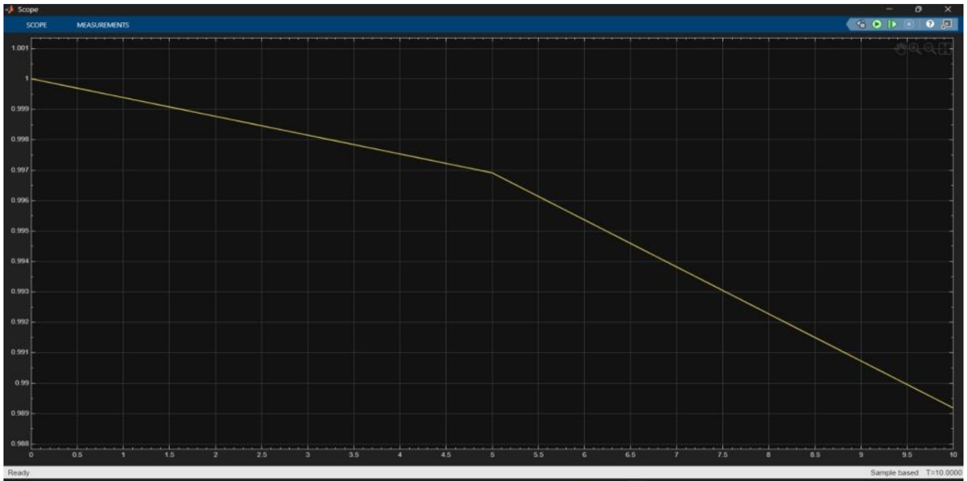

Figure 5: SOC depletion rate increases noticeably after t = 5 s — the curve steepens, indicating higher effective power loss through resistive heating.

Figure 6 — Temperature (Fault Condition)

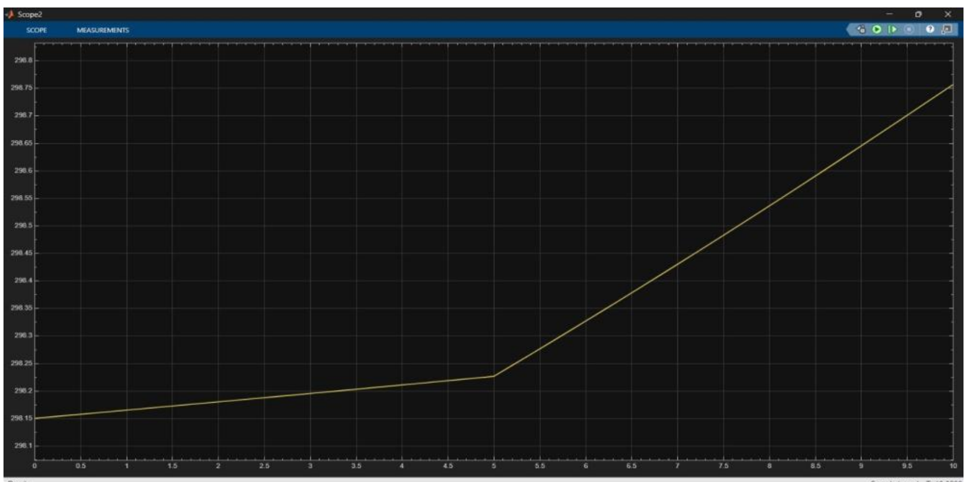

Figure 6: Temperature rises from 298.15 K at t = 0 to ~298.75 K by t = 10 s. The slope increases post-fault, evidencing escalating I^2R heat generation from the degraded cell.

Key observations during fault condition:

- Voltage Drop — the sharp step at fault time directly reflects the increased voltage loss across the higher internal resistance ($V = V_{OCV} - I \times R$).
- Accelerated SOC Depletion — additional resistive losses converted more energy to heat instead of useful output, draining charge faster.
- Thermal Escalation — temperature rise rate increased after t = 500 s, consistent with higher I^2R losses from the faulty cell.
- Fault Signature — the discontinuity in the voltage trace at t = 500 s provides a clear, detectable fault signature for real-world BMS implementations.

5.3 Post-Mitigation Condition

With the temperature-driven current limiting mitigation active, the pack was re-simulated over a longer time window (0–1000 s) to observe the stabilising effect.

Figure 7 — State of Charge (Post-Mitigation)

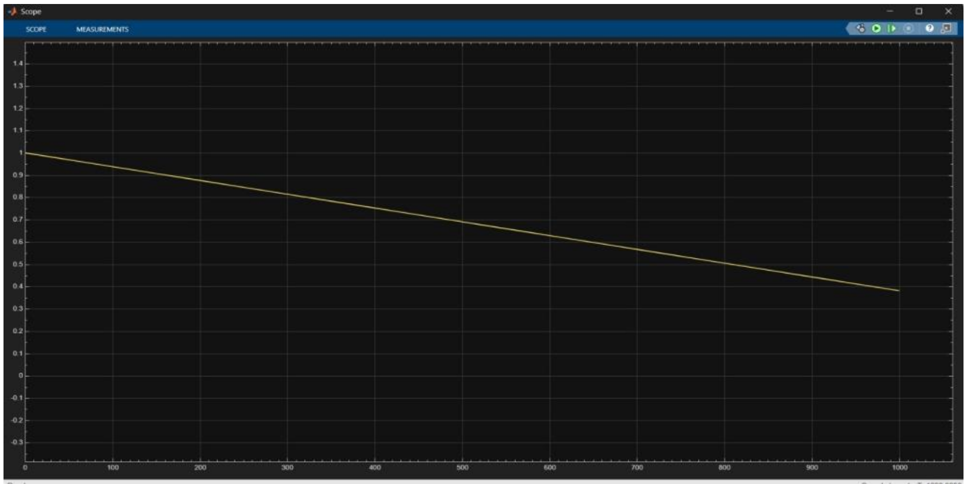

Figure 7: SOC decreases smoothly and linearly from 1.0 to ~0.38 over 1000 s — a controlled, gradual discharge with no abrupt depletion events.

Figure 8 — Temperature (Post-Mitigation)

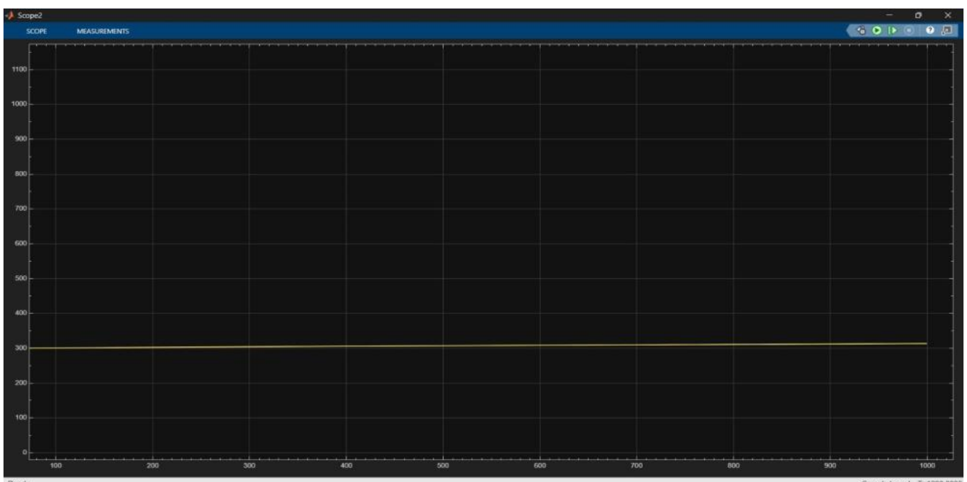

Figure 8: Temperature is maintained flat at ~300 K throughout the 1000 s simulation — the current limiting strategy completely contains thermal escalation.

Figure 9 — Terminal Voltage (Post-Mitigation)

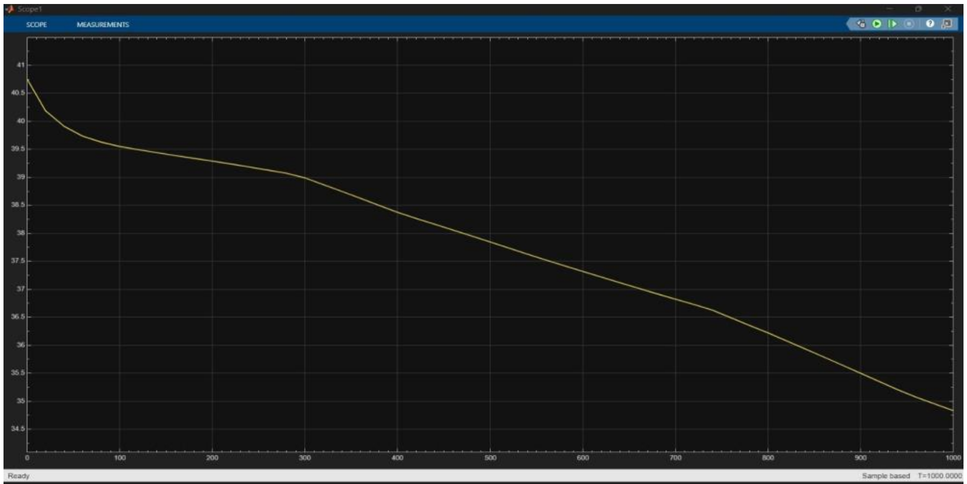

Figure 9: Voltage follows a smooth S-curve from ~40.5 V to ~34.5 V over 1000 s — a natural pack discharge profile with no fault-induced discontinuities.

Key observations post-mitigation:

- Thermal Stabilisation — temperature held constant at ~300 K throughout, confirming that the current limiting logic effectively eliminated the excess I^2R heating.
- Voltage Stability — no sharp drops observed; voltage declined smoothly along a natural discharge curve.
- Controlled SOC — discharge rate was proportional to the reduced current, extending effective pack utilisation time.
- Overall Safety — risk of thermal runaway was eliminated under the simulated fault conditions.

5.4 State of Health (SOH) — Monitoring Only

Figure 10 — State of Health (SOH)

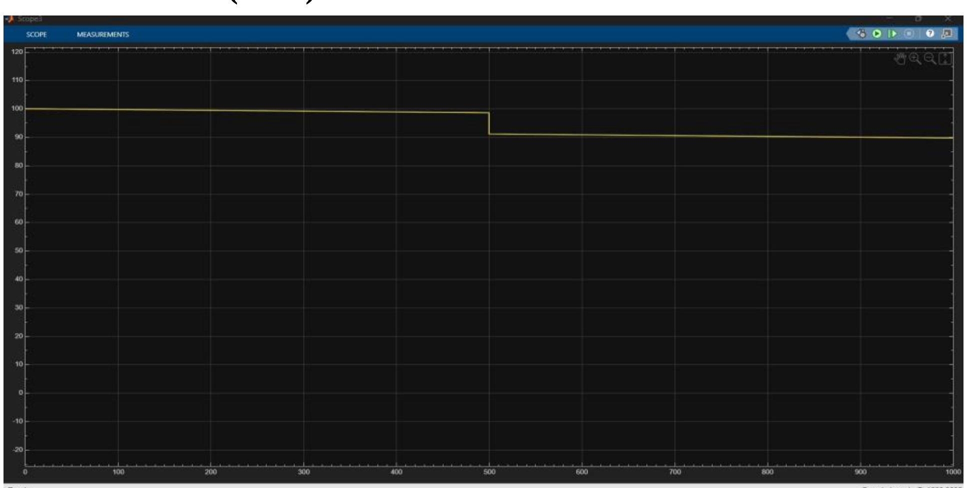

Figure 10: SOH remains at 100% during normal operation, then steps down to ~90% at t = 500 s when the fault is detected by the resistance estimator — clearly indicating cell degradation.

<u>Important clarification regarding SOH in this model:</u>

- The SOH estimator is a passive monitoring subsystem only.
- SOH output is connected exclusively to a display scope — it does not feed into any control logic, switch block, or mitigation pathway.
- The active mitigation (current limiting) is triggered by temperature alone, not by SOH.
- The SOH plot demonstrates that the estimation correctly identifies the degradation event at t = 500 s, which would be valuable in a real BMS for predictive maintenance and end-of-life decisions.
- Future implementations could integrate SOH into the control loop to pre-emptively reduce current as health degrades, before temperature thresholds are reached.

5.5 Summary Comparison Table

Parameter	Normal Operation	Fault Condition	Post-Mitigation
SOC Depletion Rate	Linear, gradual	Accelerated post-fault	Controlled, linear
Terminal Voltage	40.75 V → 40.45 V (smooth)	Sharp drop ~1.5 V at fault	40.5 V → 34.5 V (smooth)
Temperature	Stable ~300 K	Rising, accelerated post-fault	Stable ~300 K (contained)
SOH	Stable at 100%	Drops to ~90% at fault	Preserved (monitoring only)
Thermal Runaway Risk	None	Elevated	Eliminated
Pack Safety	Normal	Degraded	Restored

6. Challenges and Limitations

6.1 Challenges Encountered

- Cell Parameter Consistency — Initial model used P25B cell name in the block label while the Battery Builder was configured for the P45B cell. This was identified and corrected to ensure parameter consistency throughout.
- Fault Injection Timing — Selecting an appropriate fault injection time required balancing between showing pre-fault steady-state behaviour and post-fault response within the simulation window.
- Thermal Model Simplification — The lumped thermal model assumes uniform heat distribution across the entire pack. In reality, thermal gradients exist between cells, particularly between inner and outer cells.
- SOH Integration — Integrating the SOH estimator as a monitoring-only block (not part of the control loop) required careful attention to avoid unintended signal routing into the switching logic.
- Simulation Time Scaling — Normal discharge and mitigation simulations used different time windows (10 s vs 1000 s), requiring scope settings to be adjusted to capture meaningful behaviour in each scenario.

6.2 Limitations

- Single-Cell Fault Model — The simulation models resistance increase in the overall pack circuit rather than isolating a specific individual cell. A cell-level model with fault localisation would be more precise.
- Lumped Thermal Model — Cell-to-cell thermal coupling and propagation are not modelled. Thermal runaway propagation to adjacent cells cannot be observed in this configuration.
- Simplified SOH Estimation — The SOH formula uses fixed weightings (70% capacity, 30% resistance). A more accurate model would use empirical data from cycling experiments.
- No Physical BMS Hardware — All logic is software-only (Simulink). Real-world implementation would require MOSFET gate drivers, fuse protection hardware, CAN communication, and isolation circuits.
- Constant Current Discharge — The simulation uses constant current. Real-world EV applications exhibit dynamic, profile-based discharge patterns (WLTP, NEDC, etc.).

7. Conclusion

This project successfully demonstrated the modelling, fault injection, and mitigation of a 10s10p lithium-ion battery pack using MATLAB Simulink and Simscape Battery components. The Molicel INR-21700-P45B cell was parameterised accurately using the MATLAB Battery Builder, and the resulting pack model exhibited physically consistent behaviour across all three simulation scenarios.

The key findings of the project are:

- Internal resistance increase is an effective and realistic fault model — it produced clear, measurable signatures in voltage, SOC, and temperature traces.
- The fault injection at t = 500 s created a sharp voltage discontinuity (~1.5 V drop) and accelerated thermal rise, both of which are detectable by a real-world BMS.
- The temperature-driven current limiting mitigation strategy was highly effective — it completely stabilised pack temperature at ~300 K and restored smooth voltage and SOC profiles.
- The SOH estimator provided useful health monitoring information (dropping from 100% to ~90% at fault onset), though it was intentionally kept as a monitoring-only tool in this implementation.
- Reducing load current is a powerful thermal management technique due to the quadratic nature of Joule heating ($Q \propto I^2R$) — small current reductions yield disproportionately large reductions in heat generation.

This work lays a strong foundation for more advanced BMS designs incorporating cell-level fault localisation, multi-cell thermal propagation modelling, adaptive SOH-based current control, and hardware-in-the-loop (HIL) validation.

8. References

1. MathWorks. (2025). Simscape Battery Documentation. MATLAB R2025a. https://www.mathworks.com/help/simscape-battery/
2. Molicel. (2024). INR-21700-P45B Datasheet. https://www.molicel.com/
3. Plett, G. L. (2015). Battery Management Systems, Volume I: Battery Modeling. Artech House.
4. Saw, L. H., et al. (2016). Thermal characterisation of cylindrical lithium-ion battery cell. Applied Thermal Engineering.

9. Appendix: Git Log and Run Instructions

9.1 GitHub Repository

Repository Name: arys-battery-pack-fault-mitigation

URL: https://github.com/akshaygowdahk/arys-battery-pack-fault-mitigation

9.2 Repository Structure

Folder / File	Contents
simulation/normal_discharge/	Simulink model for baseline discharge (.slx)
simulation/fault_injection/	Simulink model with fault resistance step (.slx)
simulation/mitigation/	Simulink model with current limiting mitigation (.slx)
results/plots/	Exported scope screenshots (PNG)
results/logs/	Simulation data logs (.mat, .csv)
docs/	Report PDF and AI usage log
README.md	Project overview, cell specs, run instructions
.gitignore	MATLAB/Simulink cache and temp file exclusions

9.3 Run Instructions

Prerequisites:

- MATLAB R2025a or later
- Simscape Battery Toolbox v25.2
- Simscape Electrical Toolbox

Steps to run:

5. Open MATLAB R2025a and navigate to the simulation/ directory.
6. Open normal_discharge/ folder and launch the .slx file. Press Run. View SOC, Voltage, and Temperature scopes.

7. Open fault_injection/ folder and launch the .slx file. Press Run. Observe voltage drop and temperature rise at t = 500 s.
8. Open mitigation/ folder and launch the .slx file. Press Run. Compare temperature and voltage against fault-only results.
9. Export scope outputs via File → Export → Image (PNG) and save to results/plots/.

9.4 Incremental Git Commit Log (Expected)

Commit Message	Contents Added
Initial commit: repo structure, README, .gitignore	Folder structure, README.md, ai_usage_log.md
Add normal discharge Simulink model	simulation/normal_discharge/*.slx
Add fault injection Simulink model	simulation/fault_injection/*.slx
Add mitigation Simulink model	simulation/mitigation/*.slx
Add simulation output plots	results/plots/*.png
Add final submission report	docs/report.pdf